

NON-LINEAR DYNAMICS IN CHUA CIRCUIT

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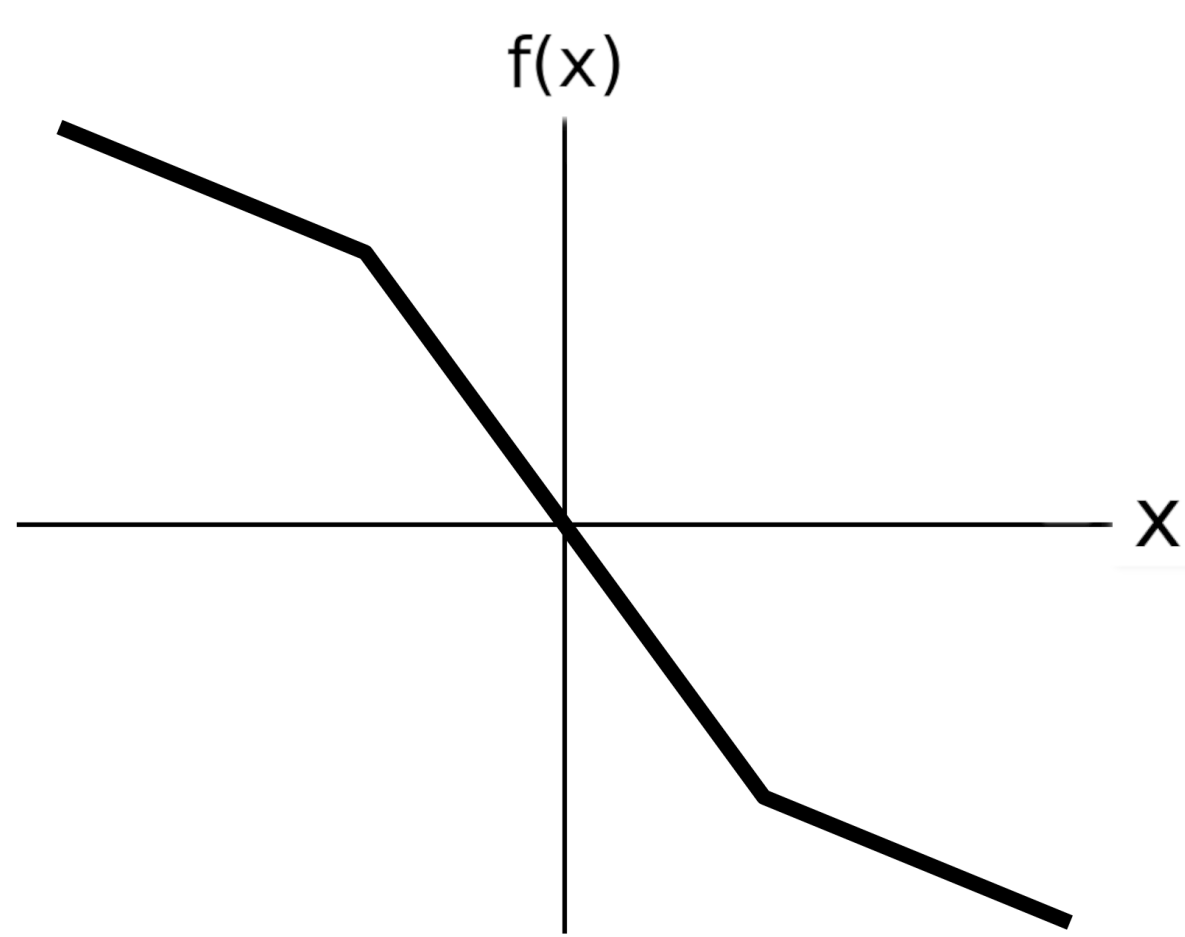


Introduction

- ✦ **Nonlinear Dynamics** is the physics that studies systems governed by equations more complex than the linear $\mathbf{aX} + \mathbf{b}$ form
- ✦ Exhibit **deterministic chaos**, where the behaviour appears **unpredictable but is not random** or unsystematic
- ✦ Electronic circuits provide **(1) a controlled setting** and **(2) rapid response times** to model nonlinear phenomena
- ✦ **Chua Circuit** is the **simplest electronic circuit** which admits chaotic behaviour
- ✦ Invented in 1983 by **Leon O. Chua** and publicized in the paper “**A chaotic attractor from chua’s circuit**” by A. Matsumoto
- ✦ Displays numerous **Bifurcations** between **Fixed Point**, **Limit Cycles** and **Strange Attractors**

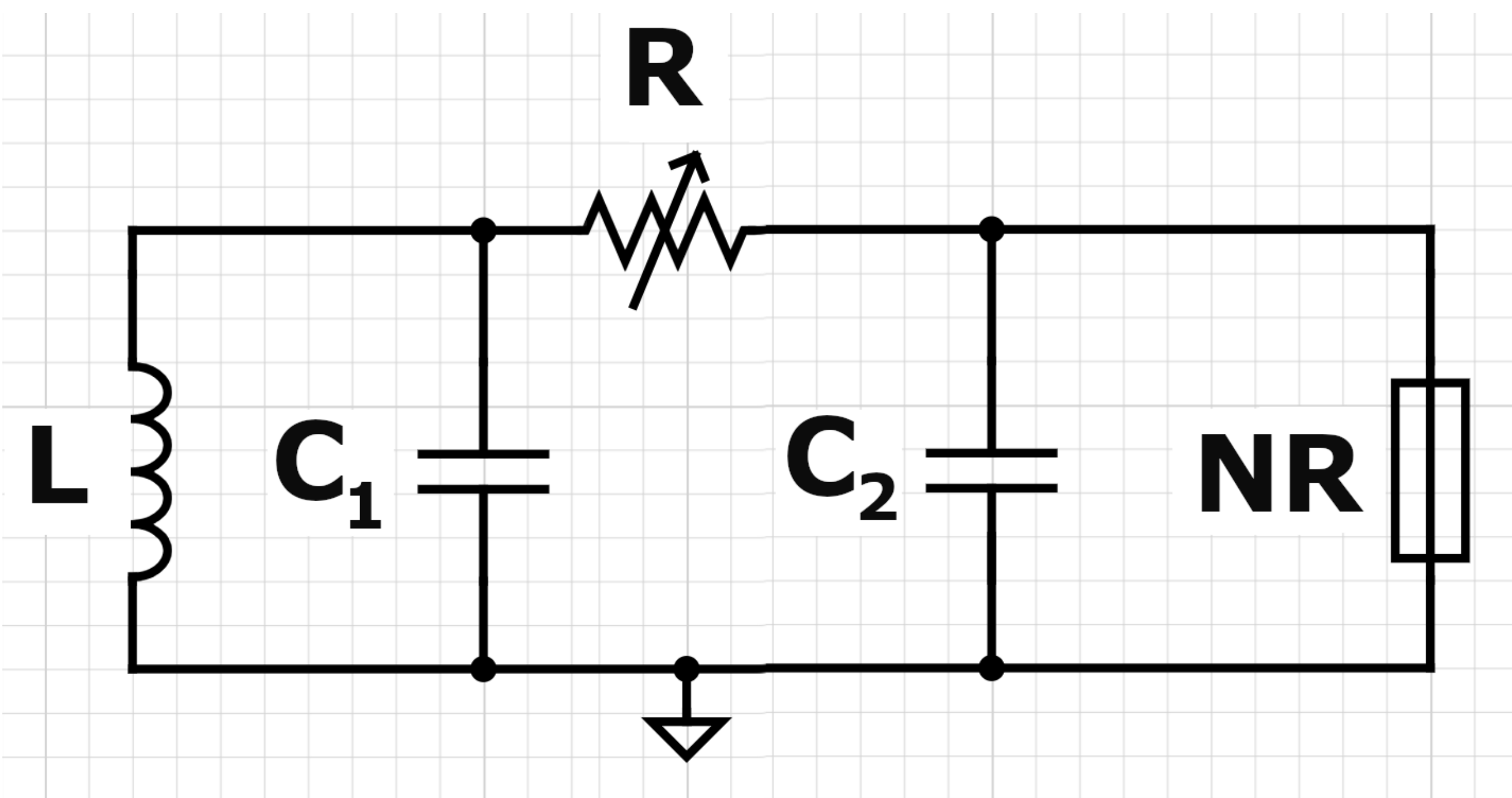
The Equations

$$\begin{aligned}\dot{x} &= \alpha[y - x - f(x)] \\ \dot{y} &= x - y + z \\ \dot{z} &= -\beta y\end{aligned}$$

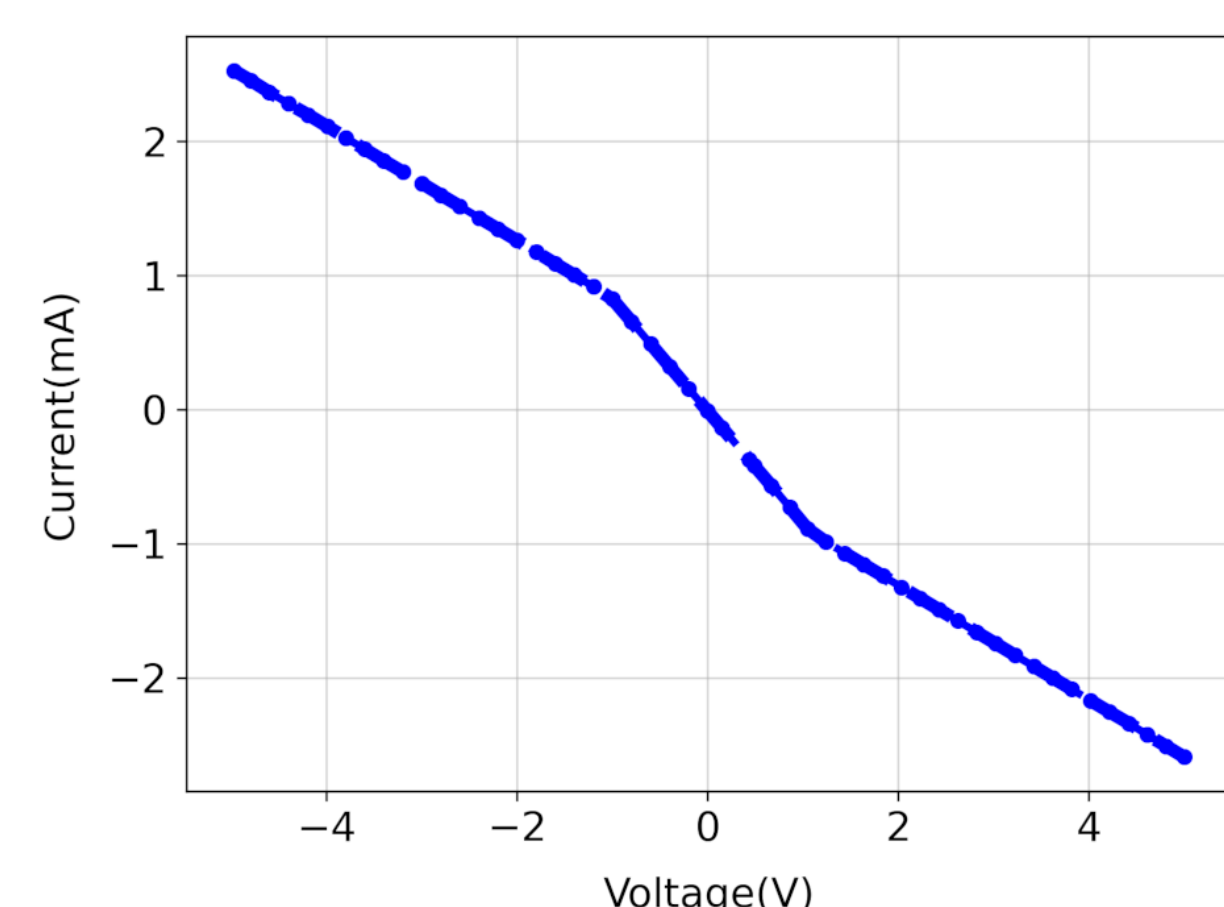
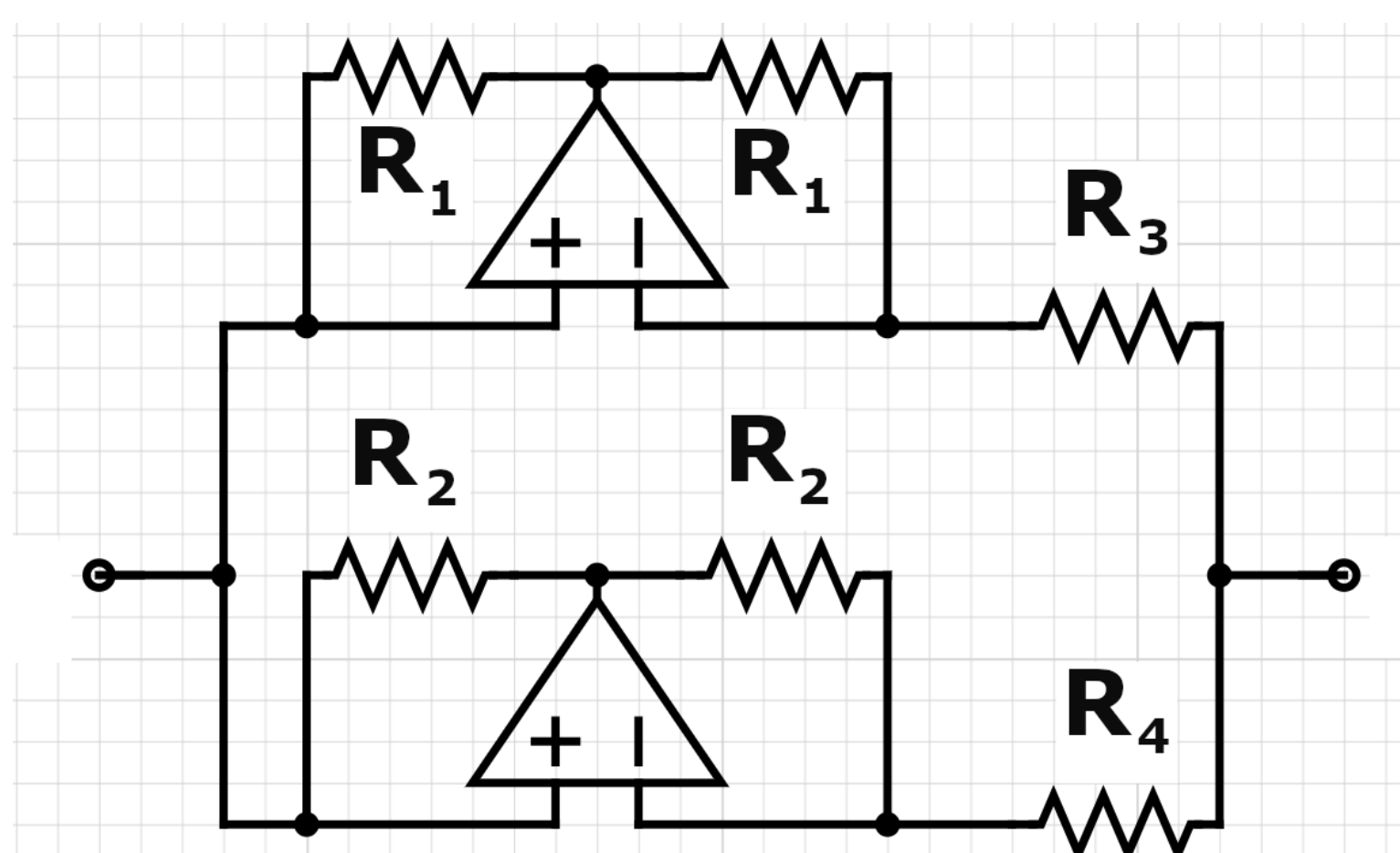


$f(x)$ is a **piecewise linear function** and acts as the nonlinear component

Circuit Diagram



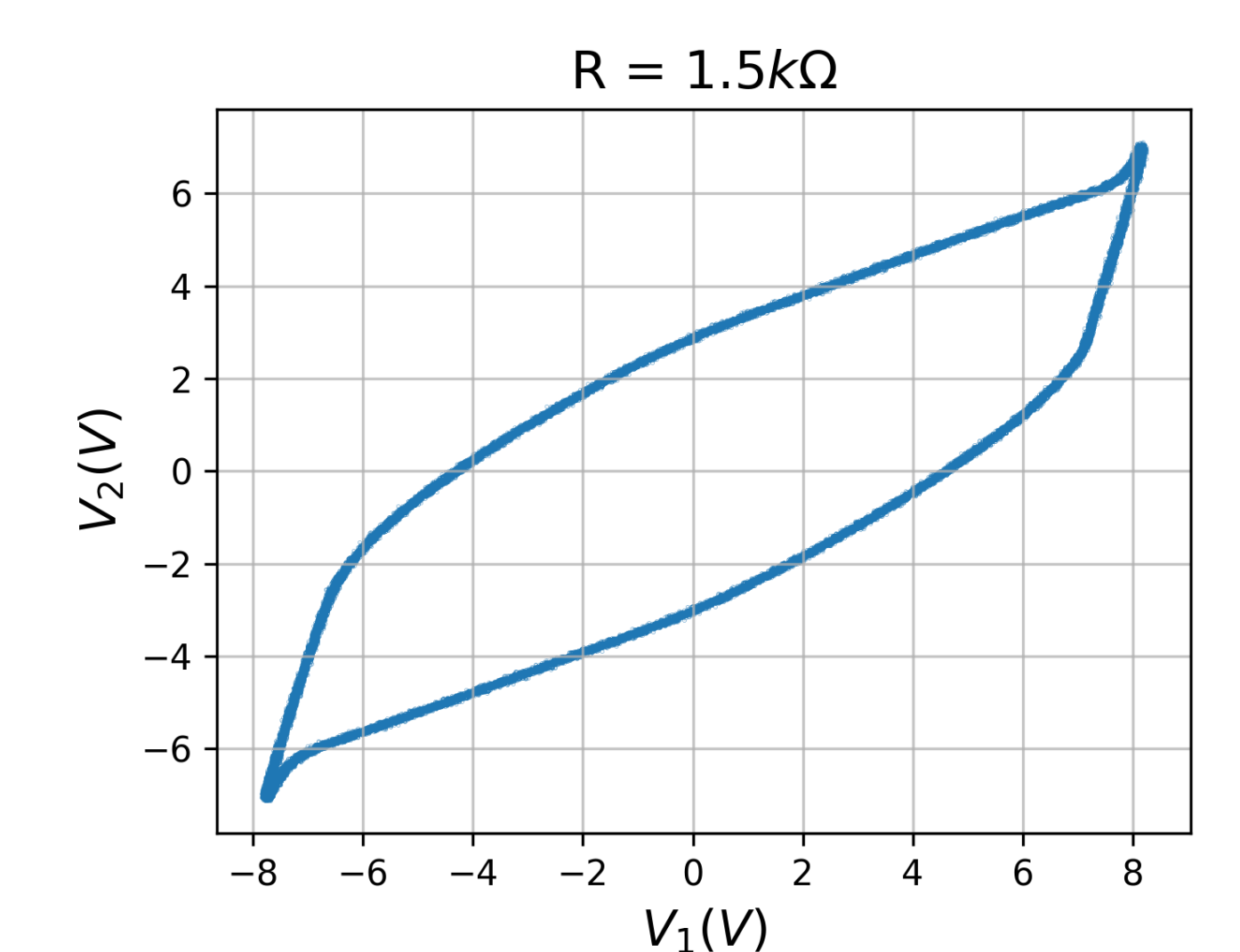
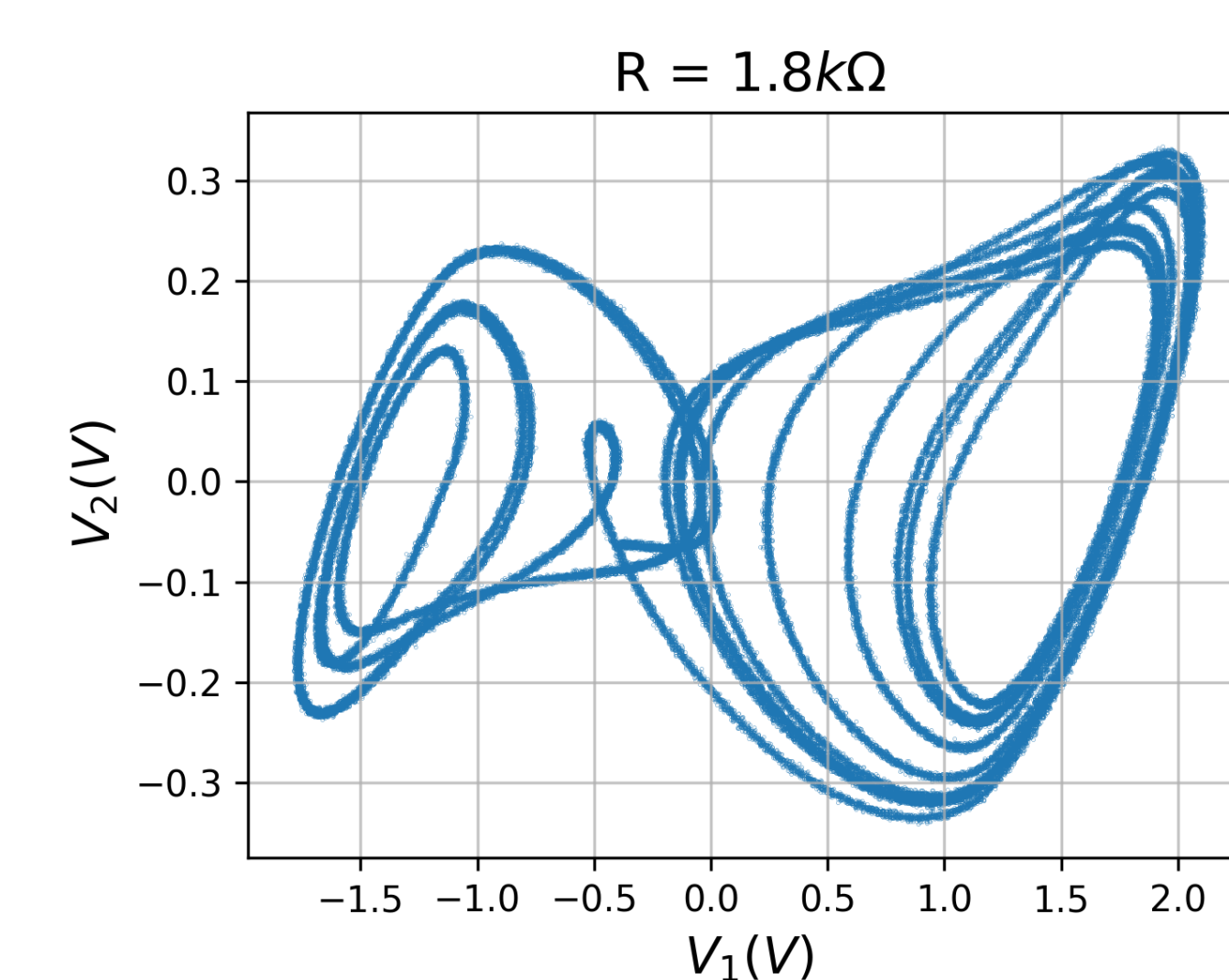
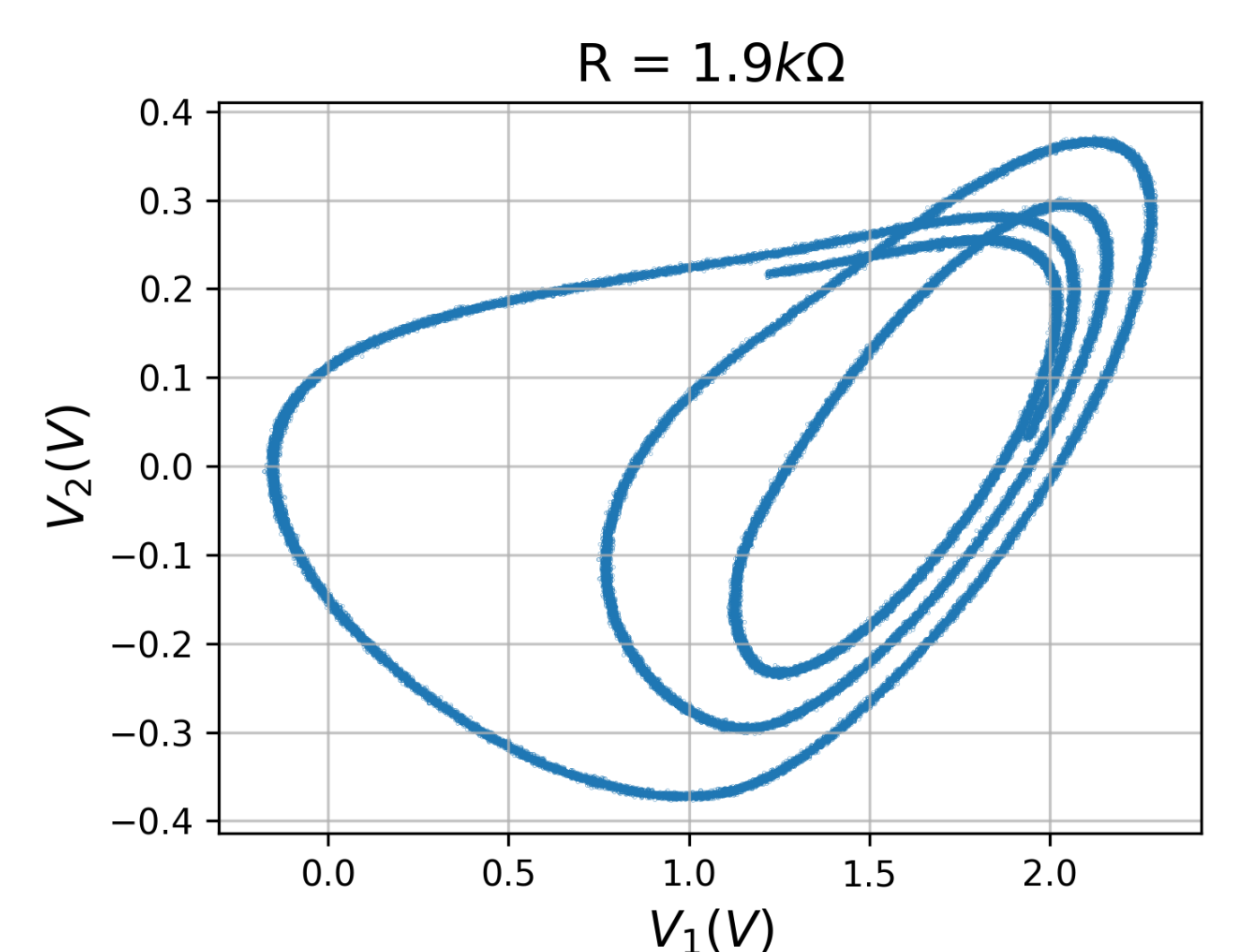
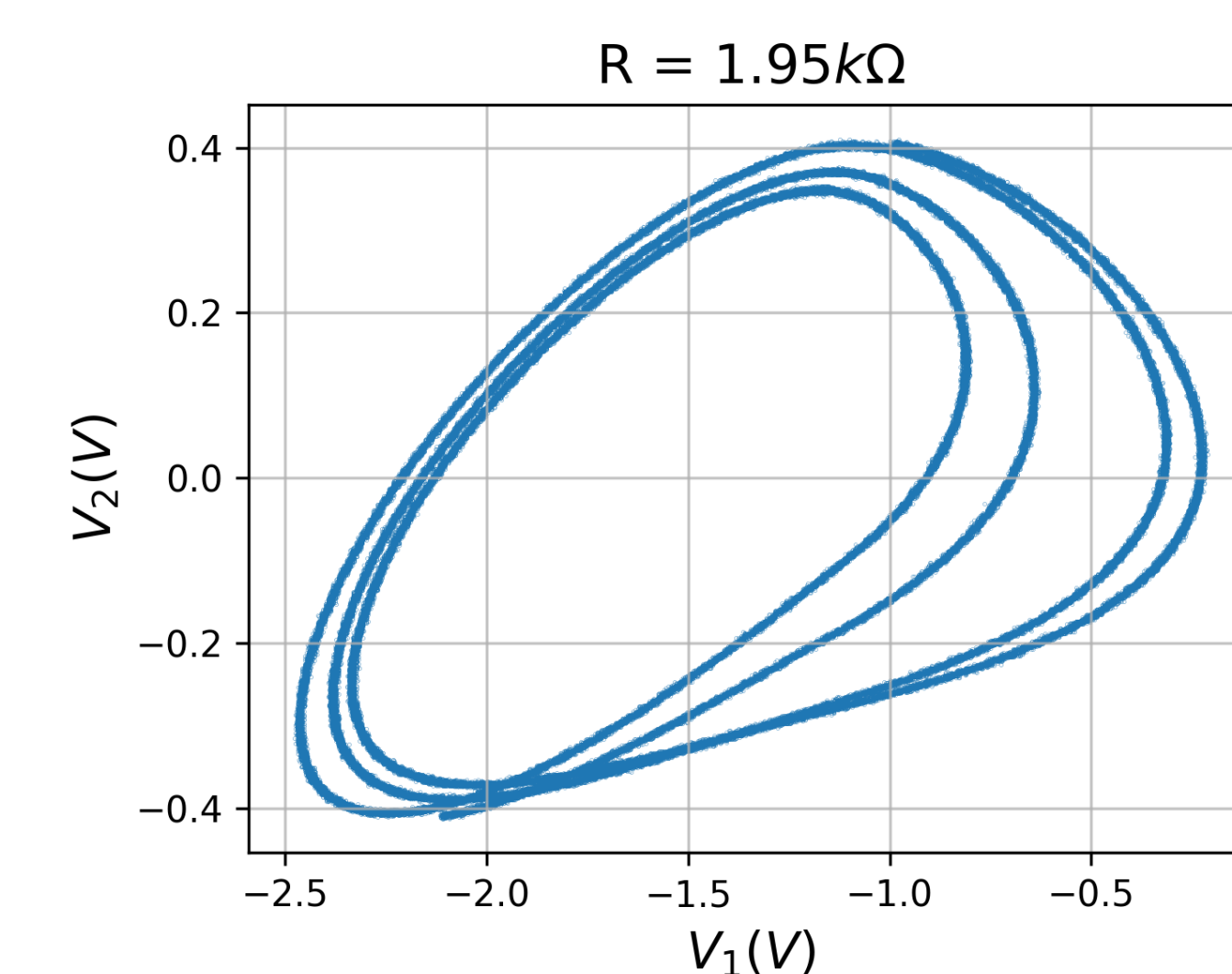
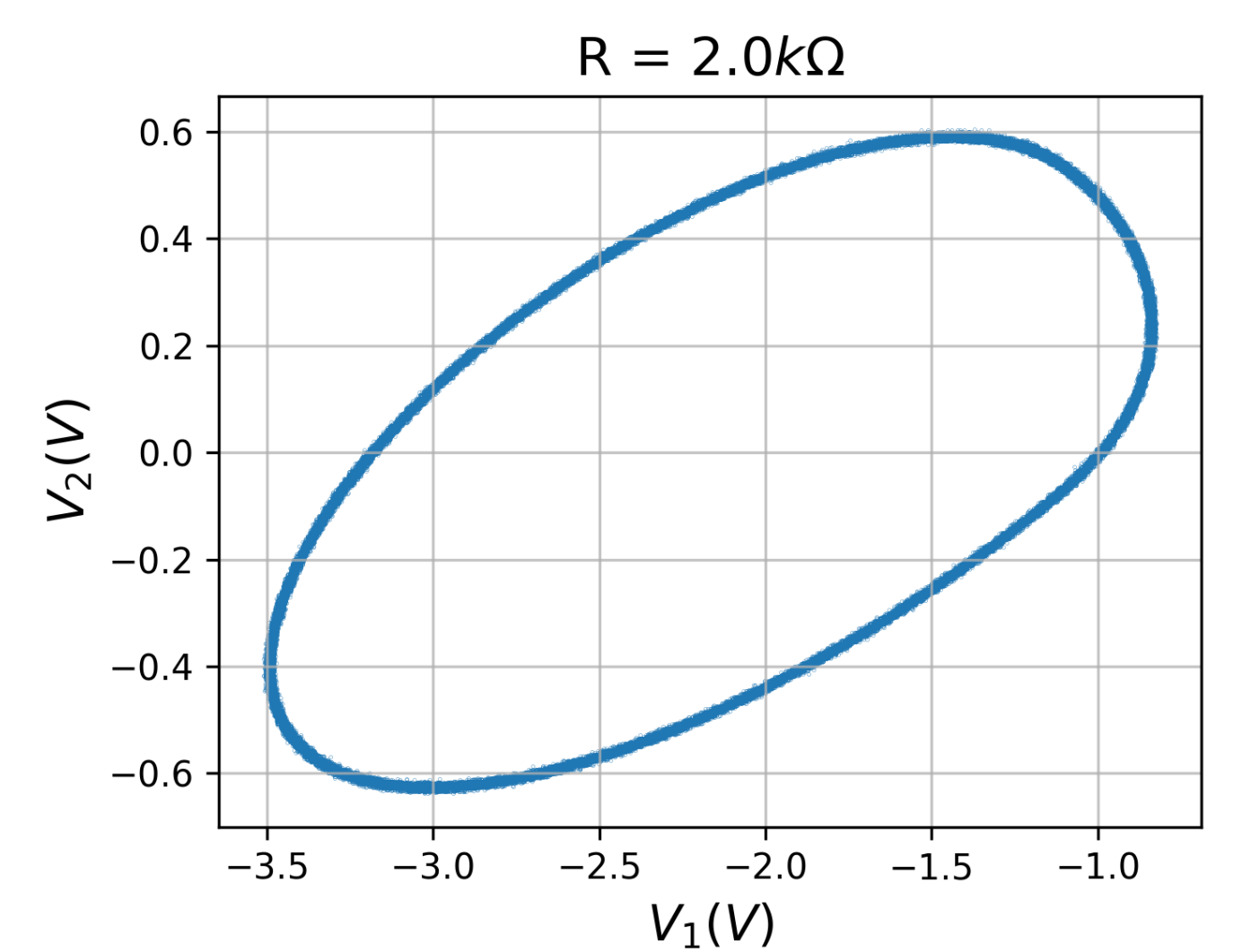
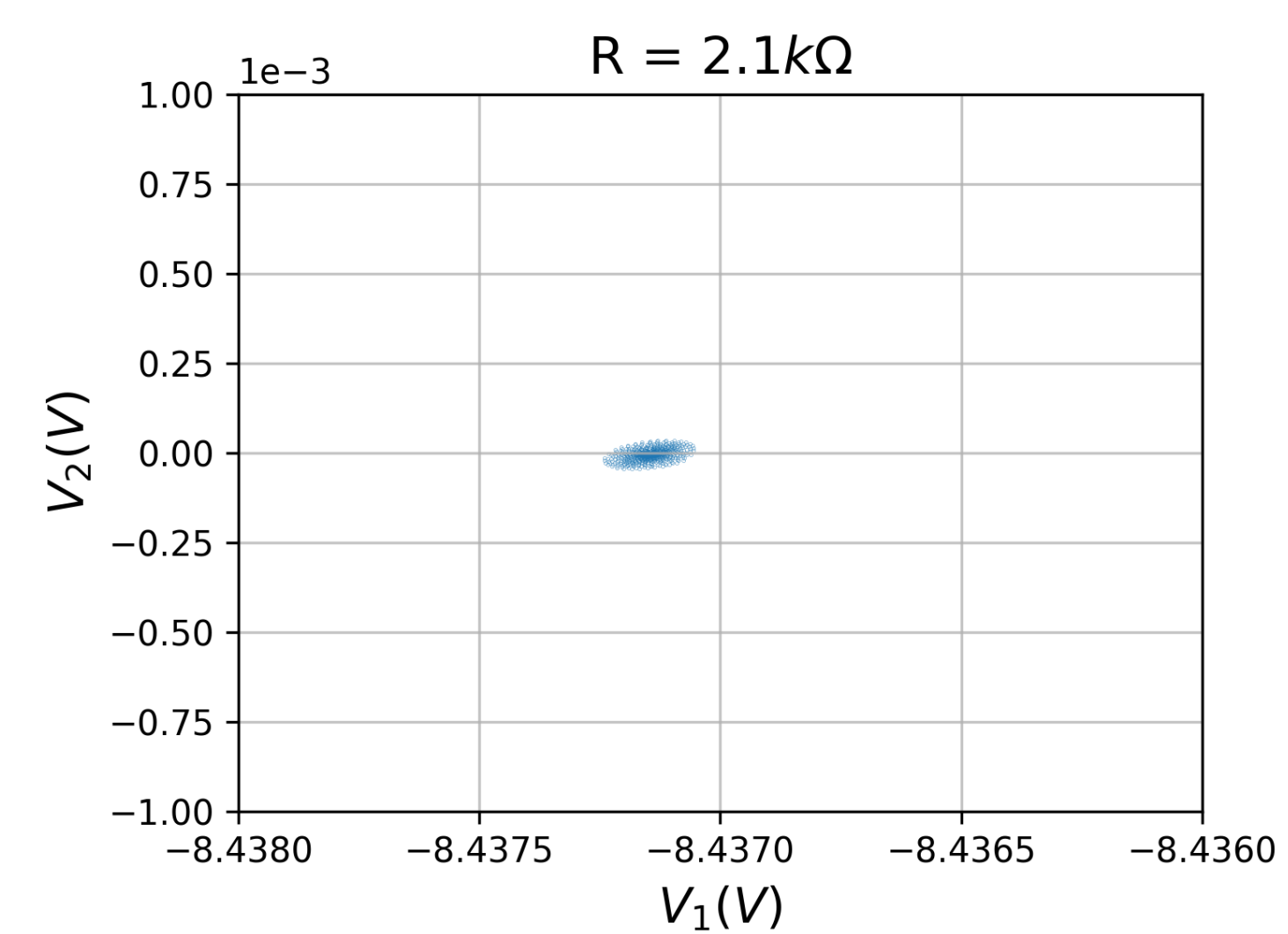
$$\begin{aligned}x &= V_{C2} & y &= V_{C1} & z &= i_L R & \tau &= t/RC_1 & f &= i_{NR} R \\ \alpha &= C_1/C_2 & \beta &= R^2 C_1/L\end{aligned}$$



Non-linear Resistor(NR) and its i-v characteristics

Results

$$\begin{aligned}L &= 18mH & C_1 &= 100nF & C_2 &= 10nF & R_1 &= 22k\Omega \\ R_2 &= 220\Omega & R_3 &= 3.3k\Omega & R_4 &= 2.2k\Omega & \text{OpAmp} &: TL082\end{aligned}$$



From left to right then top to bottom, **Stable Fixed Point**, **1-Period Limit Cycle**, **Multi-Period Limit Cycle**, **Rosler Attractor**, **Double Scroll Attractor**, **Large Limit Cycle**

Applications

- Bifurcation Analysis
- Random Number Generation
- Secure Communications
- Neural Spiking Patterns

References

- ✦ T. Matsumoto, “A chaotic attractor from chua’s circuit,” IEEE Transactions on Circuits and Systems, vol. 31, no. 12, pp. 1055–1058, 1984
- ✦ M. P. Kennedy, “Robust OP amp realization of chua’s circuit,” Frequenz, vol. 46, Jan. 1992.